

§13.5–13.7 Operators, Jacobian, IFT/ImFT

Gradient $\nabla f = (f_x, f_y, f_z)$ — points to steepest ascent, $\perp$ level sets.**Divergence** $\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = P_x + Q_y + R_z$ (scalar; net outflow).**Curl (3D)** $\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \det \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{pmatrix}$ $= (R_y - Q_z, P_z - R_x, Q_x - P_y)$.**Curl (2D)** scalar $= Q_x - P_y$. **Laplacian** $\Delta f = \operatorname{div} \nabla f = f_{xx} + f_{yy} + f_{zz}$.**Identities** $\operatorname{curl} \nabla f = \mathbf{0}$, $\operatorname{div} \operatorname{curl} \mathbf{F} = 0$.**Total deriv (Jacobian)** for $\Phi: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $D\Phi = [\partial\Phi_i/\partial x_j]$ ($m \times n$).**Chain rule** $D(\Phi \circ \Psi) = D\Phi D\Psi$ (matrix product).**Inverse Fn Thm** near $\mathbf{a}$, Φ is locally invertible $\iff \det D\Phi(\mathbf{a}) \neq 0$; then $D(\Phi^{-1}) = (D\Phi)^{-1}$.**Implicit Fn Thm** $G(\mathbf{x}, \mathbf{y}) = 0$ solves $\mathbf{y} = \mathbf{y}(\mathbf{x})$ locally if $\det(\partial G/\partial \mathbf{y}) \neq 0$; then $\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = -(G_{\mathbf{y}})^{-1} G_{\mathbf{x}}$.1-eqn: $\frac{dy}{dx} = -\frac{G_x}{G_y}$.

§14.2–14.4 Double / Triple Integrals & Fubini

Riemann $\iint_D f dA = \lim \sum f \Delta A$.**Fubini (rect)** $\iint_{[a,b] \times [c,d]} f = \int_a^b \int_c^d f dy dx = \int_c^d \int_a^b f dx dy$.**Type I** $a \leq x \leq b$, $g_1(x) \leq y \leq g_2(x)$: $\int_a^b \int_{g_1}^{g_2} f dy dx$.**Type II** $c \leq y \leq d$, $h_1(y) \leq x \leq h_2(y)$: $\int_c^d \int_{h_1}^{h_2} f dx dy$.**Swap order** sketch region $\Rightarrow$ re-read bounds the other way (a horizontal vs vertical strip). Often needed when inner antideriv is hard (e.g. e^{x^2}).**Triple** $\iiint_E f dV$; project to a plane, fill the third var first.**Volume** $= \iint_D 1 dA$ or $\iiint_E 1 dV$.

§14.6 Change of Variables (Jacobian)

2D $x = x(u, v)$, $y = y(u, v)$: $dA_{x,y} = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| dA_{u,v}$, $\frac{\partial(x,y)}{\partial(u,v)} = \det \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}$. Shortcut: $\frac{\partial(x,y)}{\partial(u,v)} = \left(\frac{\partial(u,v)}{\partial(x,y)} \right)^{-1}$.**Polar** $x = r \cos \theta$, $y = r \sin \theta$: $dA = r dr d\theta$.**Cylindrical** $dV = r dr d\theta dz$.**Spherical** $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$ $(\phi \text{ from } +z\text{-axis}, 0 \leq \phi \leq \pi)$: $dV = \rho^2 \sin \phi d\rho d\phi d\theta$.**Recipe** sub f , multiply by $|J|$, convert all bounds to new vars.

§14.8–14.9 Surface Integral (scalar) & Mass

Parametrized $S = \Phi(u, v)$ $dS = |D_u\Phi \times D_v\Phi| du dv$. $\iint_S f dS = \iint_D f(\Phi) |D_u\Phi \times D_v\Phi| du dv$.**Graph** $z = g(x, y)$ $dS = \sqrt{1 + g_x^2 + g_y^2} dA$.**Surface area** $= \iint_S 1 dS$.**Mass** $m = \int_{\Omega} \rho$ (over area dA / volume dV / surface dS).**Moments** $M_{yz} = \int x\rho$, $M_{xz} = \int y\rho$, $M_{xy} = \int z\rho$.**Center of mass** $\bar{\mathbf{r}} = \frac{1}{m} \int_{\Omega} \mathbf{r} \rho$, i.e. $\bar{x} = M_{yz}/m$, $\dots$ **Centroid** = COM with $\rho \equiv 1$: $\bar{\mathbf{r}} = \frac{1}{|\Omega|} \int_{\Omega} \mathbf{r}$.**Average value** $\bar{f} = \frac{1}{|\Omega|} \int_{\Omega} f$. **Inertia** $I = \int (\text{dist to axis})^2 \rho$, $I_z = \int (x^2 + y^2) \rho$.**Symmetry** centroid lies on any axis/plane of symmetry.

Common Solid Regions (bounds)

Disk R (polar) $0 \leq r \leq R$, $0 \leq \theta \leq 2\pi$.**Ball R (sph.)** $0 \leq \rho \leq R$, $0 \leq \phi \leq \pi$, $0 \leq \theta \leq 2\pi$.**Cylinder R, h** $0 \leq r \leq R$, $0 \leq \theta \leq 2\pi$, $0 \leq z \leq h$.**Cone** $z^2 = x^2 + y^2$, $z \geq 0$ is $\phi = \frac{\pi}{4}$; solid inside $0 \leq \phi \leq \frac{\pi}{4}$.**“Ice-cream” (cone $\cap$ sphere $\rho=1$)** $0 \leq \rho \leq 1$, $0 \leq \phi \leq \frac{\pi}{4}$, $0 \leq \theta \leq 2\pi$.**Tip** match coords to symmetry; $z^2 = x^2 + y^2 \Rightarrow \tan \phi = 1$.

§15.2 Work / Line Integral & Gradient Thm

Work $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$. scalar form $\int P dx + Q dy + R dz$.**Scalar line int** $\int_C f ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| dt$.**Gradient Thm (FTC for lines)** if $\mathbf{F} = \nabla f$ then $\int_C \nabla f \cdot d\mathbf{r} = f(\text{end}) - f(\text{start})$ (path-independent).**Conservative tests** (simply-conn. domain): $\mathbf{F} = \nabla f \iff \operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \mathbf{0}$ (curl = the **cross-product** determinant, see Ch13) $\iff$ path-indep $\iff \oint = 0$. 3D componentwise: $P_y = Q_x$, $P_z = R_x$, $Q_z = R_y$. (2D: just $P_y = Q_x$.)**Find potential f (3 components)** check $\operatorname{curl} \mathbf{F} = \mathbf{0}$ first, then:(1) $f = \int P dx + g(y, z)$; (2) set $f_y = Q \Rightarrow$ get g_y , integrate in y : $g = \int (\dots) dy + h(z)$; (3) set $f_z = R \Rightarrow$ get $h'(z)$, integrate for h . Now $\mathbf{F} = \nabla f$.**Exact form** $P dx + Q dy$ exact $\iff P_y = Q_x$.§15.5 Green's Theorem (plane, CCW ∂D)**Circulation/curl form** $\oint_{\partial D} P dx + Q dy = \iint_D (Q_x - P_y) dA$.**Flux/divergence form** $\oint_{\partial D} \mathbf{F} \cdot \mathbf{n} ds = \iint_D (P_x + Q_y) dA = \iint_D \operatorname{div} \mathbf{F} dA$.**Area** $A = \frac{1}{2} \oint_{\partial D} (x dy - y dx) = \oint x dy = -\oint y dx$.Orientation: region on the left; outward normal $\mathbf{n} = \mathbf{T} \times \mathbf{k}$ for flux.

§15.3–15.5 Flux, Divergence & Curl (Stokes)

Flux of $\mathbf{F}$ thru $S = \Phi(u, v)$ $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_D \mathbf{F}(\Phi) \cdot (D_u\Phi \times D_v\Phi) du dv$.(check orientation of $D_u\Phi \times D_v\Phi$; flip sign if needed.)**Graph** $z = g(x, y)$, **upward** $\mathbf{n} dS = (-g_x, -g_y, 1) dA$, so $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D (-Pg_x - Qg_y + R) dA$.**Divergence (Gauss) Thm** closed $S = \partial U$ outward: $\iint_{\partial U} \mathbf{F} \cdot d\mathbf{S} = \iiint_U \operatorname{div} \mathbf{F} dV$.**Curl (Stokes) Thm** ∂S oriented by RH rule with $\mathbf{n}$: $\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot d\mathbf{S}$.**Tricks** hard $\oint \Rightarrow$ pick easy capping S (Stokes); hard closed flux $\Rightarrow$ use div. If $\operatorname{curl} \mathbf{F} = \mathbf{0}$ on S , Stokes flux = 0.

§15.7–15.8 Decomposition & Differential Forms

Helmholtz / Hodge split write $\mathbf{F} = \nabla \phi + \mathbf{F}_0$ with $\operatorname{div} \mathbf{F}_0 = 0$: choose ϕ with $\Delta \phi = \operatorname{div} \mathbf{F}$ so $\nabla \phi$ is curl-free, $\mathbf{F} - \nabla \phi$ is divergence-free.**1-forms** $\omega = P dx + Q dy + R dz$; $\int_C \omega = \text{work}$.**Wedge (antisym.)** $dx \wedge dy = -dy \wedge dx$, $dx \wedge dx = 0$.So $\int z^2 dy \wedge dx = -\int z^2 dx \wedge dy$ (a flux/area 2-form).**d of forms** $d(P dx + Q dy) = (Q_x - P_y) dx \wedge dy$ (Green).

Worked: potential & Green

Conservative $\mathbf{F} = (2xy, x^2)$: $P_y = 2x = Q_x \checkmark$. $f = \int 2xy dx = x^2 y + g(y)$; $f_y = x^2 + g' = x^2 \Rightarrow g \text{ const.}$ So $\int_A^B \mathbf{F} \cdot d\mathbf{r} = [x^2 y]_A^B$.**Green area** $\oint_{\partial D} (-y dx + x dy) = 2 \text{Area}(D)$.**Stokes shortcut** replace messy $\mathbf{F}$ by simpler $\mathbf{G}$ with same curl on S — both give equal $\oint$.

§16.2–16.3 Series Convergence Tests

 n th-term if $a_n \not\rightarrow 0 \Rightarrow$ diverges (necessary, not suff.).**Geometric** $\sum ar^n = \frac{a}{1-r}$ if $|r| < 1$, else div. **p -series** $\sum 1/n^p$ conv $\iff p > 1$.**Integral** $a_n = f(n) \geq 0 \downarrow$: $\sum a_n$ & $\int_1^\infty f$ together.**Comparison** $0 \leq a_n \leq b_n$: $\sum b_n$ conv $\Rightarrow \sum a_n$ conv.**Limit comp.** $\lim a_n/b_n = L \in (0, \infty) \Rightarrow$ same behavior.**Ratio** $L = \lim \left| \frac{a_{n+1}}{a_n} \right|$: $L < 1$ conv (abs), $L > 1$ div, $L = 1$?**Root** $L = \lim |a_n|^{1/n}$: same rule as ratio.**Alternating (Leibniz)** $\sum (-1)^n b_n$, $b_n \downarrow 0 \Rightarrow$ conv; error $\leq b_{N+1}$.**Abs vs cond** $\sum |a_n|$ conv $\Rightarrow$ absolutely; if $\sum a_n$ conv but $\sum |a_n|$ div $\Rightarrow$ conditionally (rearrangeable to any sum, Riemann).**Dirichlet** $b_n \downarrow 0$, partial sums of $\sum c_n$ bounded $\Rightarrow \sum b_n c_n$ conv.**Pick a test** factorial / $n^n \rightarrow$ ratio or root; rational in $n \rightarrow$ limit-compare to p -series (leading powers); alternating $\rightarrow$ Leibniz, then check $\sum |a_n|$.

§17.3–17.4 Power Series & Taylor

$\sum a_n(x-c)^n$ radius R : $\frac{1}{R} = \lim \left| \frac{a_{n+1}}{a_n} \right| = \lim |a_n|^{1/n}$.
Converges $|x-c| < R$, diverges $> R$; **test endpoints separately**.
Interval $= (c-R, c+R) \pm$ endpoints (use other tests there).
Term-by-term differentiate/integrate inside R (same R).
Taylor $f(x) = \sum \frac{f^{(n)}(c)}{n!} (x-c)^n$.
Maclaurin $e^x = \sum \frac{x^n}{n!}$, $\sin x = \sum \frac{(-1)^n x^{2n+1}}{(2n+1)!}$, $\cos x = \sum \frac{(-1)^n x^{2n}}{(2n)!}$,
 $\frac{1}{1-x} = \sum x^n$, $\ln(1+x) = \sum \frac{(-1)^{n-1} x^n}{n}$.
More binomial $(1+x)^k = \sum \binom{k}{n} x^n$ ($|x| < 1$); $\frac{1}{(1-x)^2} = \sum n x^{n-1}$.
Integrate/diff term-by-term plug into a known series, then $\int$ or $\frac{d}{dx}$ (same R). Ex $\frac{1}{1+x^2} = \sum (-1)^n x^{2n} \Rightarrow \arctan x = \int = \sum \frac{(-1)^n x^{2n+1}}{2n+1}$;
 $\int e^{x^2} dx = \sum \frac{x^{2n+1}}{n!(2n+1)} + C$.

§18.1–18.3 Fourier Series

2π -periodic $f \sim a_0 + \sum_{n \geq 1} (a_n \cos nx + b_n \sin nx)$:
 $a_0 = \frac{1}{2\pi} \int_0^{2\pi} f dx$ (the mean),
 $a_n = \frac{1}{\pi} \int_0^{2\pi} f \cos nx dx$, $b_n = \frac{1}{\pi} \int_0^{2\pi} f \sin nx dx$.
Period $2L$ replace $nx \rightarrow \frac{n\pi x}{L}$, $\frac{1}{L} \rightarrow \frac{1}{L}$, $\frac{1}{2\pi} \rightarrow \frac{1}{2L}$.
Symmetry f even $\Rightarrow b_n=0$ (cosine series); f odd $\Rightarrow a_n=0$ (sine series).
Convergence (Dirichlet) at a jump the series $\rightarrow \frac{f(x^+) + f(x^-)}{2}$ (midpoint); to f where continuous.
Complex form $f = \sum_{n \in \mathbb{Z}} c_n e^{inx}$, $c_n = \frac{1}{2\pi} \int_0^{2\pi} f e^{-inx} dx$.
Parseval $\frac{1}{2\pi} \int_0^{2\pi} |f|^2 = \sum |c_n|^2$; real form $\frac{1}{\pi} \int_0^{2\pi} f^2 = 2a_0^2 + \sum (a_n^2 + b_n^2)$.
Half-range $[0, L]$ even ext. $\rightarrow$ cosine $a_n = \frac{2}{L} \int_0^L f \cos \frac{n\pi x}{L}$; odd ext. $\rightarrow$ sine $b_n = \frac{2}{L} \int_0^L f \sin \frac{n\pi x}{L}$.
Parseval trick sawtooth $b_n = \frac{2}{n}$: $\sum \frac{4}{n^2} = \frac{1}{\pi} \int_0^{2\pi} (\pi-x)^2 = \frac{2\pi^2}{3} \Rightarrow \sum \frac{1}{n^2} = \frac{\pi^2}{6}$.

Common Parametrizations & Normals

Circle R $\mathbf{r}(t) = (R \cos t, R \sin t)$, $|\mathbf{r}'| = R$.
Sphere R $\Phi = (R \sin \phi \cos \theta, R \sin \phi \sin \theta, R \cos \phi)$; $D_\phi \Phi \times D_\theta \Phi$ outward, $|\cdot| = R^2 \sin \phi$; outward $\mathbf{n} = \frac{1}{R}(x, y, z)$.
Cylinder R $\Phi = (R \cos \theta, R \sin \theta, z)$; normal $(\cos \theta, \sin \theta, 0)$, $|\cdot| = R$.
Graph $z = g(x, y)$ $\Phi = (x, y, g)$; $D_x \Phi \times D_y \Phi = (-g_x, -g_y, 1)$ (upward).
Plane thru $\mathbf{r}_0$ spanned by $\mathbf{a}, \mathbf{b}$: $\mathbf{n} = \mathbf{a} \times \mathbf{b}$.

Vector Identities (product rules)

$\nabla(fg) = f \nabla g + g \nabla f$.
 $\operatorname{div}(f\mathbf{F}) = f \operatorname{div} \mathbf{F} + \nabla f \cdot \mathbf{F}$.
 $\operatorname{curl}(f\mathbf{F}) = f \operatorname{curl} \mathbf{F} + \nabla f \times \mathbf{F}$.
 $\operatorname{div}(\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot \operatorname{curl} \mathbf{F} - \mathbf{F} \cdot \operatorname{curl} \mathbf{G}$.
 $\operatorname{curl}(\operatorname{curl} \mathbf{F}) = \nabla(\operatorname{div} \mathbf{F}) - \Delta \mathbf{F}$.

Integration Toolbox

Half-angle $\sin^2 x = \frac{1-\cos 2x}{2}$, $\cos^2 x = \frac{1+\cos 2x}{2}$.
Orthogonality on $[0, 2\pi]$ ($m, n \geq 1$):
 $\int_0^{2\pi} \cos mx \cos nx = \int_0^{2\pi} \sin mx \sin nx = \pi \delta_{mn}$;
 $\int_0^{2\pi} \sin mx \cos nx = 0$; $\int_0^{2\pi} \sin^2 = \int_0^{2\pi} \cos^2 = \pi$.
By parts $\int x \sin nx dx = \frac{\sin nx}{n^2} - \frac{x \cos nx}{n}$;
 $\int x \cos nx dx = \frac{\cos nx}{n^2} + \frac{x \sin nx}{n}$.
Gaussian $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ (via $\iint e^{-r^2} r dr d\theta$).

Notable Sums (endpoint sanity checks)

$\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$, $\sum_{n \geq 1} \frac{1}{n^4} = \frac{\pi^4}{90}$.
 $\sum_{n \geq 1} \frac{(-1)^{n-1}}{n} = \ln 2$, $\sum_{n \geq 0} \frac{(-1)^n}{2n+1} = \frac{\pi}{4}$.
Harmonic $\sum 1/n$ diverges; alternating harmonic conv. (conditionally).

Worked Fourier: sawtooth $f(x) = \pi - x$ on $[0, 2\pi]$

Odd about its mean $\Rightarrow a_0 = 0$, $a_n = 0$.
 $b_n = \frac{1}{\pi} \int_0^{2\pi} (\pi - x) \sin nx dx = \frac{2}{n}$. $S(x) = \sum_{n \geq 1} \frac{2}{n} \sin nx$.
At $x = 0$ (jump $\pi \rightarrow -\pi$): $S(0) = \frac{f(0^+) + f(0^-)}{2} = \frac{\pi + (-\pi)}{2} = 0$.

Which theorem? (decision guide)

$\oint_C \mathbf{F} \cdot d\mathbf{r}$, C in plane $\rightarrow$ **Green**. $\oint_C \mathbf{F} \cdot d\mathbf{r}$, C in space $\rightarrow$ **Stokes** (cap S).
 $\iint_{\partial U} \mathbf{F} \cdot d\mathbf{S}$ closed $\rightarrow$ **Divergence**. $\mathbf{F} = \nabla f$ / $\operatorname{curl} = 0 \rightarrow$ **Gradient Thm** (endpoints). Open surface flux $\rightarrow$ parametrize Φ , use $\mathbf{F} \cdot (D_u \Phi \times D_v \Phi)$.

§13.4 Newton's Method (root-finding)

Core (solve $\mathbf{F}(\mathbf{x}) = 0$) $\mathbf{x}_{n+1} = \mathbf{x}_n - [D\mathbf{F}(\mathbf{x}_n)]^{-1} \mathbf{F}(\mathbf{x}_n)$.
1D $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ (tangent's x -axis hit; quadratic conv.).
2x2 inverse $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
Linear approx $\mathbf{F}(\mathbf{x}_0 + \mathbf{h}) \approx \mathbf{F}(\mathbf{x}_0) + [D\mathbf{F}(\mathbf{x}_0)]\mathbf{h}$ (differentiable if all partials continuous).
To solve $\mathbf{F}(\mathbf{x}) = \mathbf{b}$ apply to $\mathbf{F} - \mathbf{b}$: $\mathbf{x}_{n+1} = \mathbf{x}_n - [D\mathbf{F}]^{-1}(\mathbf{F}(\mathbf{x}_n) - \mathbf{b})$.
Ex $\sqrt{2}$ $f = x^2 - 2 \Rightarrow x_{n+1} = \frac{1}{2}(x_n + \frac{2}{x_n})$: $1 \rightarrow 1.5 \rightarrow 1.41\bar{6} \rightarrow 1.41422$.

Shortcuts that save time

Symmetry $\Rightarrow 0$ $\int$ of an *odd* integrand over a symmetric domain $= 0$ (apply componentwise to flux/work/centroid).
Constant div $\operatorname{div} \mathbf{F} = c \Rightarrow$ flux through closed $S = c \cdot \operatorname{Vol}(U)$. Const. field $\mathbf{F} = \mathbf{c}$: $\operatorname{div} = 0$.
Curl is source-free $\iint_{\partial U} (\operatorname{curl} \mathbf{F}) \cdot d\mathbf{S} = 0$ (closed surface, always).
Conservative $\operatorname{curl} \mathbf{F} = \mathbf{0} \Rightarrow \oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ on any closed C (simply-conn.).
Useful constants sphere R : area $4\pi R^2$, vol $\frac{4}{3}\pi R^3$; $\int_0^{2\pi} d\theta = 2\pi$, $\int_0^\pi \sin \phi d\phi = 2$.
Area as line int $A = \frac{1}{2} \oint (x dy - y dx)$.

Tangent Plane & Surface Normal

Level surface $F(x, y, z) = c$ normal $= \nabla F$; tangent plane $\nabla F(\mathbf{p}) \cdot (\mathbf{r} - \mathbf{p}) = 0$.
Graph $z = f(x, y)$ plane $z = f(a, b) + f_x(x-a) + f_y(y-b)$; normal $(-f_x, -f_y, 1)$.
Linear approx $f(\mathbf{p} + \mathbf{h}) \approx f(\mathbf{p}) + \nabla f(\mathbf{p}) \cdot \mathbf{h}$.

Orientation conventions

Green (2D) ∂D counterclockwise, region on your left.
Stokes orient ∂S by RH rule with $\mathbf{n}$: thumb $= \mathbf{n}$, fingers curl along ∂S (surface on left).
Divergence $\mathbf{n}$ points **outward** from the solid U .
Split boundary $\partial S = \gamma + \eta \Rightarrow \oint_{\partial S} = \oint_\gamma + \oint_\eta$; reversing a curve flips its sign.

Quadric Surfaces (recognize fast)

Paraboloid $z = x^2 + y^2$ (bowl). **Cone** $z^2 = x^2 + y^2$.
Sphere $x^2 + y^2 + z^2 = R^2$. **Cylinder** $x^2 + y^2 = R^2$.
Ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.
Hyperboloid $x^2 + y^2 - z^2 = 1$ (1 sheet), $z^2 - x^2 - y^2 = 1$ (2 sheets).

The four theorems = one family

$\int_{\partial \Omega} \omega = \int_{\Omega} d\omega$ — boundary integral = integral of a derivative over the region:
FTC $\int_a^b f' = f|_a^b$; Gradient $\int_C \nabla f \cdot d\mathbf{r} = f|_{\text{ends}}$; Green/Stokes $\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S}$; Divergence $\iint_{\partial U} \mathbf{F} \cdot d\mathbf{S} = \iiint_U \operatorname{div} \mathbf{F} dV$.

Exam playbook (from 2025 final)

1. Iterated int $\rightarrow$ sketch region, swap order, Fubini. 2. Green: change-of-var Jacobian, work + flux, split curl/div-free. 3. Stokes: $\operatorname{curl} \mathbf{F}$, cap with disk, get $\oint$. 4. Gauss + spherical on "ice-cream" cone $\cap$ sphere; forms $dy \wedge dx$. 5. Torus Φ : normal dir, centroid of S^+ , flux, $\iiint |z|$. 6. Series: cond/abs, rearrangement $S \neq s$, radius R , Fourier a_n, b_n .